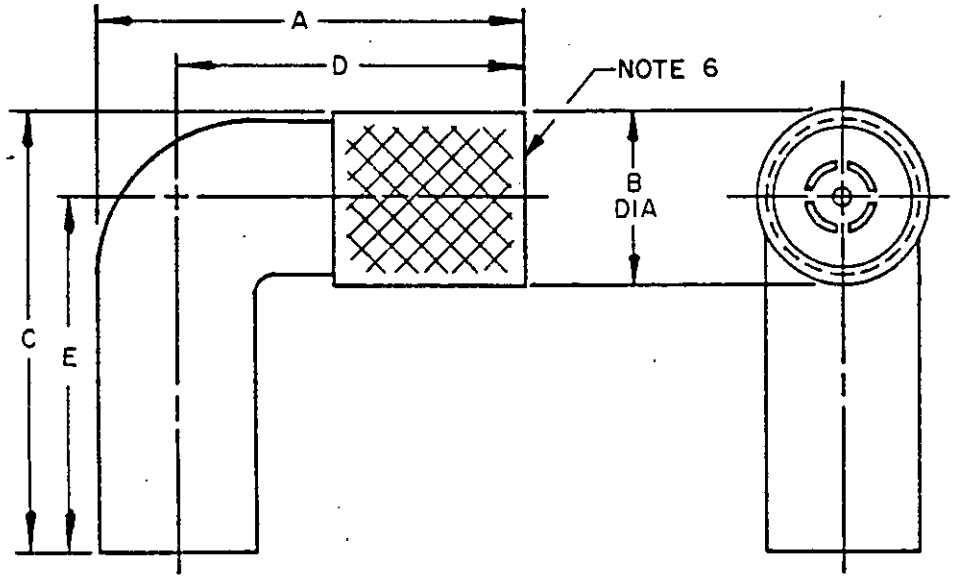
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES N (CABLED), RIGHT ANGLE, PIN CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



MARKING IMPLEMENTATION DATE,
CATEGORY B, SEE TABLE IV

NOTES:
1. See table I and III for dimensions A, B, C, D and E.
2. Dimensions A, B and C are the largest overall dimensions of the connector.
3. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
4. All undimensioned pictorial representations are for reference purposes only.
5. Dimension C defines the maximum length of the connector when assembled to the appropriate cable.
6. Series N, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A
FSC 5935
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TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Applicable cable M17/ #	Dimensions	Inches (millimeters) Minimum	Maximum
Category A - Field serviceable (no special tools required)				
0101 4/	6-RG11Δ 62-RG144Δ 065-RG165 074-RG213 75-RG214* 77-RG216Δ RG225/U*	A B C	1.312(33.32)	1.500(38.10) .827(21.01) 2.370(58.42)
0006 4/	2-RG6Δ 073-RG212* RG222/U			
Category C - Field replaceable (MIL-C-22520 Crimp tool) 3/ See footnote symbol next to applicable cable for crimp die				
0004 4/	074-RG213*<			
0005 4/	75-RG214*<			
0013 4/	2-RG6*Δ≠			
0014 4/	6-RG11Δ< 62-RG144*@Δ<	A	1.312(33.32)	1.531(38.89)
0015 4/	065-RG165*@<	B C		.827(21.010 2.375(60.33)
0016 4/	073-RG212*≠ RG-222/U≠			
0017 4/	77-RG216Δ*<			
0018 4/	RG-225/U*@<			
Category D - Field replaceable - defined piece parts 3/5/2/				
0501 4/	75-RG214* RG-225/U@ 127-RG393	A		1.406(35.72)
0502 4/	074-RG213*	B C	1.312(33.32)	.827(21.01) 2.218(56.36)
0503 4/	060-RG142*@ 128-RG400	D E		1.078(27.38) 1.812(46.04)
0504 4/	028-RG058			

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

3/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 3.

4/ These connectors have captivated center contacts.

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5/ Not to be used in Army equipment.

\# The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and ∆.

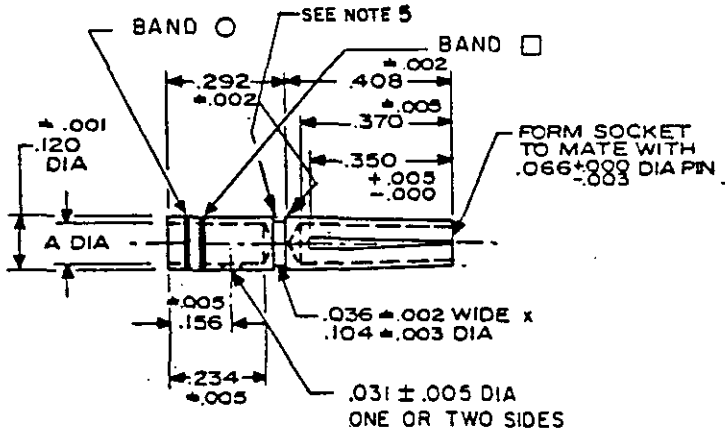
@ Cable to be used for the +200°C temperature cycling tests.

∆ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

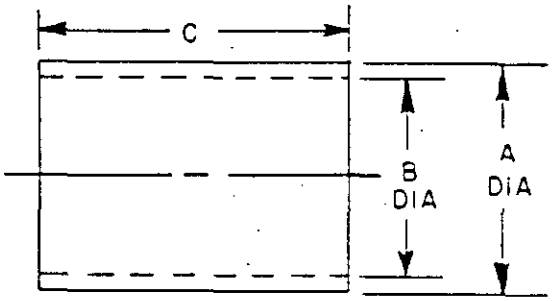
≠ M22520/5-35 Closure A or M22520/5-55 Closure A

≤ M22520/5-61

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CENTER CONTACT



CRIMP FERRULE

Dash No.	Contact No. 2/	A	Basic crimp tool 1/ M22520/1	Crimp die or positioner M22520/1	Crimp tensile min	Color band □	Color band ○
0501 0502	5-10	0.098 ±0.002	-01	-13	60 lbs	Red	Orange
0503 0504	5-11	0.043 +0.001 -0.002	-01	-13	20 lbs	BLUE	Orange

Dash No.	Ferrule No. 2/	A	B	C ±0.015	Basic crimp tool 1/ M22520/5	Crimp tool M22520/5
0501	5-50	0.492 +0.003 -0.000	0.438 +0.003 -0.000	0.600	-01	-25 Closure A or 61
0502	5-51		0.418 +0.003 -0.000			
0503	5-52	0.250 ±.003	0.220 ±.003	0.500	-01	5, 11 or 57 Closure A or 19 Closure B
0504	5-53	.245 ±.001	.206 ±.001	0.500		

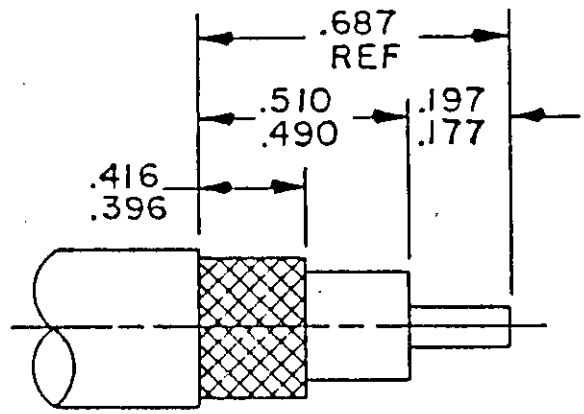
Inches	mm	Inches	mm
.001	0.03	.156	3.96
.002	0.05	.220	5.59
.003	0.08	.234	5.94
.005	0.13	.250	6.35
.015	0.38	.292	7.42
.031	0.79	.350	8.89
.036	0.91	.370	9.40
.043	1.09	.408	10.36
.066	1.68	.418	10.62
.098	2.49	.438	11.13
.104	2.64	.492	12.50
		.500	12.70
		.600	15.24

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Crimp tensile test shall be in accordance with MIL-C-39029.
4. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
5. .003 maximum break.

FIGURE 2. Contact and ferrule dimensions for category D only.

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Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Recommended cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

1,000 volts rms, maximum working voltage at sea level.

250 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque: 6 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds, minimum.

Inspection conditions: Coupling torque, 6 to 10 inch-pounds.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Outer contact:

Test ring I. D.: .316 maximum, 16 microinch finish.

Insertion force: 25 pounds, maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .324, minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only). Fifteen pounds, minimum axial force for all cable except RG400 and RG142; 6 pounds, minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower.

From 500 to 9,000 MHz: 1.35, maximum.

From 9,000 to 11,000 MHz: 1.50, maximum.

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Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.008+.002 F (F in GHz).

Item 16: VSWR shall be less than 1.008+.002 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.012+.004 F (F in GHz).

Group B inspection: VSWR shall be less than 1.024+.007 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.08.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	2.5	3.0
Outer contact	.2	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 2,500 volts rms, minimum at sea level for connectors using other than RG400 and RG142; 1,500 volts rms, minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 500 volts, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,500 volts rms, tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 75 pounds, minimum.

Crimp assemblies:

50 pounds, minimum for cables .155 - .189 OD.

60 pounds, minimum for cables .190 - .229 OD.

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75 pounds, minimum for cables .230 - .249 OD.

90 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz. This requirement may be met by qualifying the straight connector version for the same cable under MIL-C-39012/1.

Insertion loss: .30 dB, maximum tested at 10 GHz.

Part number: M39012/05 (dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/05	M39012/05
I	-0101	-0101
II	-0006	-0006
III	B0002 B0003	B0002 B0003
IV	-0004 -0005 -0015 -0016 -0018	-0004 -0005 -0015 -0016 -0018
V	-0013 -0014 -0017	-0013 -0014 -0017
VI	B0007 B0008 B0011	B0007 B0008 B0011
VII	B0009 B0010 B0012	B0009 B0010 B0012
VIII	-0501 -0502	-0501 -0502 -0503
IX	-0503	-0503
X	-0504	-0504

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

Dash number 1/ 3/	Applicable cable M17/ #	Dimensions	Inches (millimeters) Minimum	Maximum
M39012/05B0002 4/ 1/	074-RG213*			
M39012/05B0003 4/	75-RG214*			
M39012/05B0007 2/	2-RG6Δ			
M39012/05B0008 2/	6-RG11Δ 62-RG144*@Δ	A	1.312(33.32)	1.531(38.89)
M39012/05B0009 2/	065-RG165*@	B C		.827(21.01) 2.375(60.33)
M39012/05B0010 2/	073-RG212* RG222/U			
M39012/05B0011 2/	77-RG216*Δ			
M39012/05B0012 2/	RG225/U*@			

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ These connectors have captivated center contacts.

3/ For maintenance replacements for category B, see table V.

4/ Inactive for new design.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C temperature cycling tests.

Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

Preferred part number M39012/05	Substitute for part number or type designation 1/2/
-0101	UG-594C/U, M39012/05-0001
B0002	UG-1707/U, M39012/05-0002
B0003	UG-1708/U, M39012/05-0003
-0004	
-0005	
-0006	
B0007	M39012/05-0007
B0008	M39012/05-0008
B0009	M39012/05-0009
B0010	M39012/05-0010
B0011	M39012/05-0011
B0012	M39012/05-0012
-0013	
-0014	
-0015	
-0016	
-0017	
-0018	
-0501	
-0502	
-0503	
-0504	

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/05-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

Category B number*	Category C dash number	Category A dash number	Category D dash number
B0002	0004	0101	0502
B0003	0005	0101	0501
B0007	0013	0006	---
B0008	0014	0101	---
B0009	0015	0101	---
B0010	0016	0006	---
B0011	0017	0101	---
B0012	0018	0101	0501

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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